American Computer Science League

2023-2024 ● Contest 1: Short Problems Solutions ● Senior Division

1. Computer Number Systems

Subtract the 2 ticket numbers and add 1.
$F4A_{16} - 3A5_{16} = BA5_{16}$
$BA5_{16} + 1_{16} = BA6_{16}$
Converting BA6 to base 10 is 11*256+10*16+6 = 2560+256+160+6=2982.

A. 2982

2. Computer Number Systems

$7245_8 * 17_8 = 7245_8 * (20-1)_8$
$= 165120_8 - 7245_8$
$= 155653_8$

B. 155653

3. Recursive Functions

$f(15) = f(15-2) \bmod 4 + 2 = f(13) \bmod 4 + 2$
$= 4 \bmod 4 + 2 = 0 + 2 = 2$
$f(13) = f(13-2) \bmod 4 + 2 = f(11) \bmod 4 + 2$
$= 2 \bmod 4 + 2 = 2 + 2 = 4$
$f(11) = f(11-2) \bmod 4 + 2 = f(9) \bmod 4 + 2$
$= 8 \bmod 4 + 2 = 0 + 2 = 2$
$f(9) = f(9-1) \bmod 5 + 4 = f(8) \bmod 5 + 4$
$= 4 \bmod 5 + 4 = 4 + 4 = 8$
$f(8) = f(8-1) \bmod 5 + 4 = f(7) \bmod 5 + 4$
$= 5 \bmod 5 + 4 = 0 + 4 = 4$
$f(7) = f(7-1) \bmod 5 + 4 = f(6) \bmod 5 + 4$
$= 11 \bmod 5 + 4 = 1 + 4 = 5$
$f(6) = 6 + 5 = 11$
So $f(f(f(15))) = f(f(2)) = f(2+5) = f(7) = 5$

C. 5

4. Recursive Functions

There are 8 32" squares that are subdivided. Each of those has 8 32/3" squares. Each of those has 8 32/9" squares. Each of those has 9 32/27" squares. These are smaller than 2" so the recursion stops. Therefore, there are a total of 8*8*8*9 = 4608 small squares.

C. 4608

5. What Does This Program Do? The year 2023 is the only year other than 2400 that is the product of the sum of its digits and the square of the sum of the squares of its digits. In other words, (2+0+2+3)*(2^2+0^2+2^2+3^2)^2 = 7*17^2 = 7*289 = 2023. Its prime factors also are these two sums since 7*17*17 is its prime factorization.	C. 2023